



Multiple-reaction kinetics of composite electrodes for sulfide-based all-solid-state batteries: Impedance decoupling

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ABSTRACT

All-solid-state batteries (ASSBs) have garnered considerable attention as next-generation energy-storage systems owing to their safety. In ASSBs, composite electrodes comprising solid electrolytes, active materials, and conductive additives are employed to increase the electrochemically active surface and ensure the Li^+ /electron pathways. Composite electrodes consist of numerous solid-solid contact interfaces and exhibit interfacial side reactions that inhibit charge transfer. Thus, a comprehensive analysis of their electrochemical processes is necessary to develop high-performance ASSBs. Herein, we demonstrate the quantitative analyses of the multiple-reaction kinetics in a solid composite electrode for ASSBs with state-of-the-art $\text{Li}_6\text{PS}_5\text{Cl}$ electrolytes and $\text{LiNi}_{0.8}\text{Co}_{0.1}\text{Mn}_{0.1}\text{O}_2$ active materials using modified transmission line models based on networks of distributed impedance elements. Firstly, model simulations are conducted with different boundary conditions and circuit elements to investigate the impedance responses of electrodes. The circuit elements represent the interfacial and charge/atom transport processes, namely, Li^+ /electron transport, charge transfer, double-layer charging, and diffusion in the active material. Secondly, for quantitative experimental analyses, two model cells (i.e., symmetric cells and full cells) are fabricated with various electrode formulations, and their impedance spectra are measured as a function of temperature. Kinetic parameters are determined using the appropriate equivalent circuit models to elucidate the effects of electrode formulation and operating conditions on electrochemical performance. Finally, the interplay between the kinetics of ionic transport and interfacial reaction is discussed based on measurements using uncoated active materials and halide electrolytes. Thus, this study affords insights into the electrochemical processes of ASSBs and guidelines for quantitative analyses via the impedance decoupling method.

1. Introduction

The rapid growth of the electric vehicle (EV) market has increased the demand for high-energy-density secondary batteries. Conventional Li-ion cells have been used for EV battery packs, offering high operation voltage and cycling stability [1,2]. However, the use of Li-ion batteries is limited by safety issues concerning the use of flammable organic liquid electrolytes [3,4]. To address this issue, all-solid-state batteries (ASSBs) have emerged as the ultimate candidate for next-generation battery systems owing to their intrinsic safety [5,6]. Among solid electrolytes, sulfide-based electrolytes are widely used for ASSBs because they offer high ionic conductivity and have favorable mechanical properties [7,8]. Their intrinsic deformability enables plastic deformation under applied pressure without fracture, allowing densification of pellets by simple

cold pressing. This densification promotes intimate particle-particle contact and reduces grain-boundary resistance. These unique mechanical characteristics make sulfide electrolytes particularly suitable for roll-to-roll dry electrode processes, where powders can be uniformly distributed and densified through solvent-free compression. The dry electrode process provides advantages such as increased electrode loading and homogeneous microstructures without crack formation or binder aggregation [9,10].

In conventional Li-ion batteries with liquid electrolytes, the electrolyte easily wets the electrode laminate, facilitating charge transfer at the active material/electrolyte interfaces. Meanwhile, in ASSBs, composite electrodes consisting of solid electrolytes, active materials, and conductive additives are commonly employed to enlarge the electroactive area and secure the Li^+ /electron pathways [11,12]. These

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electrodes contain numerous interfaces, which are entirely composed of solid–solid contacts and thus make the reaction kinetics sluggish. Consequently, various elemental reactions should be all considered for electrochemical analysis, including Li^+ /electron transport, charge transfer, double-layer charging, and diffusion in the active material [13–15]. Furthermore, complicated degradation processes—such as reduction/oxidation of the solid electrolyte, physical contact loss due to repeated volume change during charge–discharge cycles, and chemical side reactions that form the resistive layer—increase the interfacial resistance and limit the electrochemical performance of ASSBs [16,17]. Therefore, a comprehensive understanding of the electrochemical processes in composite electrodes is necessary to develop high-performance ASSBs.

Among various methods, electrochemical impedance spectroscopy (EIS) is a powerful nondestructive tool for deconvolution of complex electrochemical processes, which measures the impedance of a system over a wide frequency range [18,19]. Physical phenomena with different kinetics and, therefore, different time constants can be distinguished and quantitatively evaluated using EIS. The impedance response of a system with liquid electrolytes is generally analyzed using the Randles circuit or its modified versions [20–22]. This simplified interpretation is facilitated by rapid Li^+ /electron transport in conventional Li-ion batteries with liquid electrolytes. However, for composite electrodes with solid electrolytes, charge carriers experience high transport resistance, prompting the need to consider Li^+ /electron pathways for ASSBs [18,23,24]. The charge transport in composite electrodes can be described using the transmission line model (TLM), which considers different pathways for different charge carriers [25–28]. The TLM is characterized by a ladder-like structure composed of an infinite number of infinitesimal circuit elements distributed along the thickness direction. In the TLM, charge carriers (electrons and Li^+) move through different rails, and they interact at the crosswise rail between two rails, which is analogous to the electroactive interface. Consequently, TLMs can decouple multiple processes, making it a useful tool for describing the impedance response of solid composite electrodes.

In this study, we constructed TLMs to describe the electrochemical processes of ASSB composite electrodes and demonstrated the decoupling of their kinetics for parameter extraction. Two boundary conditions were considered in the equivalent circuit modeling, namely, ion blocking–ion blocking and electron blocking–ion blocking, where each circuit element corresponded to a specific electrochemical process, including charge transport, interfacial charge transfer, double-layer charging, and diffusion in the active material. Furthermore, two different types of model cells, i.e., symmetric cells and full cells with sulfide electrolytes, were fabricated with various electrode formulations and tested under different operating conditions. Subsequently, the impedance responses of the composite electrodes were quantitatively analyzed to deconvolute complicated electrochemical processes and elucidate the critical factors that determine electrochemical performance.

2. Experimental

2.1. Fabrication of composite cathodes

Composite cathodes were fabricated by mixing $\text{LiNi}_{0.8}\text{Co}_{0.1}\text{Mn}_{0.1}\text{O}_2$ (NCM811, COSMO AM&T), $\text{Li}_6\text{PS}_5\text{Cl}$ (LPSCl, POSCO-JK Solid Solution), and Super P using a ball mill with zirconia balls at 250 rpm for 48 h. The weight ratios of NCM811, LPSCl, and Super P were set to 62:37:1, 72:27:1, and 82:17:1, respectively. For the fabrication of halide-based composite cathodes, Li_2ZrCl_6 (LZC) powder was prepared via high-energy ball milling using a planetary micro-mill (Pulverisette 7, Fritsch). Stoichiometric amounts of LiCl (Alfa Aesar) and ZrCl_4 (Sigma Aldrich) were ball-milled at 700 rpm for 12 h. The weight ratio of NCM811, LZC, and Super P was fixed at 72:27:1 for the halide-based

composite cathodes.

2.2. Composite cathode symmetric cell for electrochemical analysis

Composite cathode pellets (0.2 mm thickness and 10 mm diameter) were fabricated by cold pressing the powder samples. Specifically, 40 mg of composite cathode powder was pressed under 250 MPa for 90 s. Subsequently, stainless-steel (SUS) foils (thickness of 10 μm) were placed on both sides of the composite cathode pellet as current collectors. After impedance measurement, the cells were disassembled, and their thicknesses were measured in an Ar-filled glove box.

2.3. Three-electrode full cell with Ag–Li alloy reference electrode for electrochemical analysis

A Teflon-coated Ag wire (SME KOREA, diameter of 0.16 mm) was used to fabricate the reference electrode. The Teflon coating on one end of the wire was peeled off to a length of 1 mm to serve as the active area of the reference electrode. Three-electrode cells were assembled in a polyaryletheretherketone (PEEK) mold with a 0.5 mm diameter hole. LPSCl powder (100 mg) was added to the PEEK mold and gently pressed. The prepared Ag wire was inserted into the solid electrolyte with its active region positioned at the center. Subsequently, the solid electrolyte layer was pelletized at 100 MPa. A composite cathode mixture (10 mg) was pressed on one side of the solid electrolyte at 433 MPa, and In (100 μm thickness and 10 mm diameter) and Li (200 μm thickness and 4 mm diameter) foils were placed on the other side. After inserting SUS foils (10 μm thickness) as current collectors on both sides, the cell was pressed under 250 MPa for 3 min.

2.4. Electrochemical tests

The three-electrode full cells were subjected to galvanostatic cycling in the voltage range of 2.5–4.3 V vs. Li/Li^+ at 30 °C. The initial cycling was performed at 0.05C under constant-current/constant-voltage (CC–CV) charging and constant-current (CC) discharging. The cut-off current in the constant-voltage mode was set to one-fifth of the applied current in the CC mode. After the initial cycle, the cells were charged and discharged under CC–CV charging and CC discharging at 0.5C. The EIS tests were conducted using a potentiostat (Biologic SP-300) with the measuring temperature set to 30, 45, and 60 °C in a constant-temperature chamber. The EIS measurements were performed in the frequency range of 7 MHz to 50 mHz with a perturbation amplitude of 5 mV.

3. Results and discussion

3.1. Equivalent circuit modeling

Electrochemical processes in an ASSB composite electrode are illustrated in Fig. 1. The processes can be classified into ion transport, electron transport, and interfacial reaction coupled with solid-state diffusion [27,29]. The composite electrode is a mixed conductor with ionic/electronic transport contributions to its impedance. Electrons move through active material and conductive additive particles, while Li^+ ions are mainly transported through solid electrolyte particles. Charge transfer occurs at the interface between the solid electrolyte and active material particles. In addition to these processes, various electrochemical and chemical reactions occur, affecting charge transport in the composite electrode [23]. For example, since composite electrodes are typically fabricated by cold pressing, grain boundaries are formed between solid electrolyte particles. The capacitance and resistance associated with these grain boundaries can limit interparticle Li^+ transport. Moreover, solid-state (chemical) diffusion of Li in the active material affects its utilization [30,31].

To investigate these polarization processes, equivalent circuit

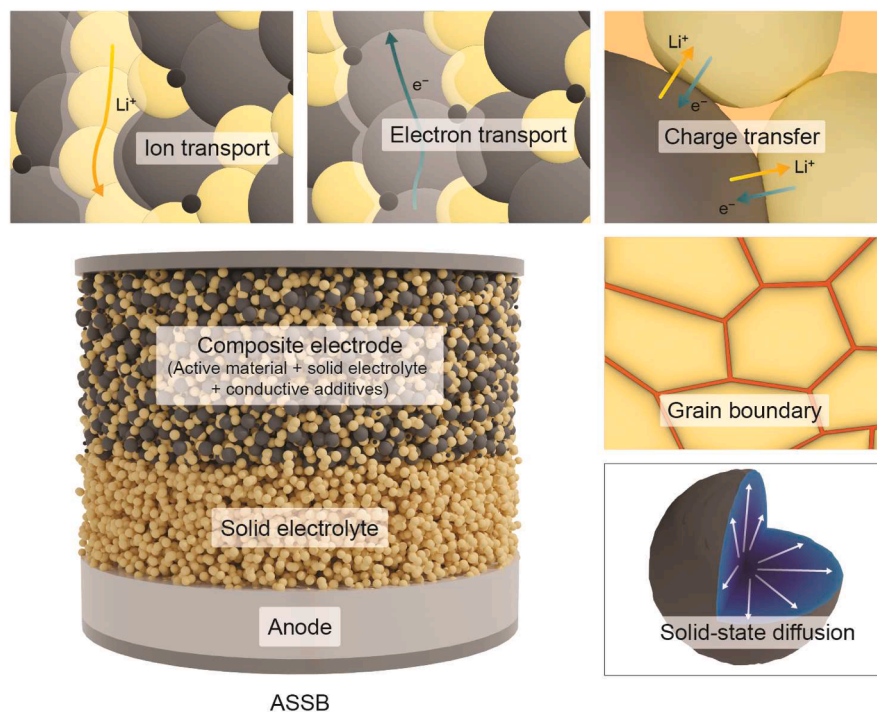


Fig. 1. Schematic of electrochemical processes in ASSB composite electrodes.

modeling was performed using the modified TLMs [27,32]. The TLMs for simulating composite electrodes comprise two rails, representing the Li^+ and electron pathways. The crosswise rail represents the impedance at the interface between solid electrolyte and active material particles. The impedance of each rail is denoted by z . Specifically, z_1 and z_3 refer to the ionic and electronic impedances per unit surface area, respectively, and z_2 is the interfacial impedance per unit volume. The detailed numerical expressions for each circuit element and total impedance of the TLM are provided in Supplementary Note 1. The frequency range was set from 7 MHz to 5 mHz, and the exponent of the constant-phase element (CPE) was fixed to 1 in the simulation.

Fig. 2a illustrates the TLM with the ion blocking–ion blocking boundary condition, in which no charge transfer occurs. The TLM corresponds to a symmetric cell with the configuration [current collector | composite electrode | current collector]. Fig. 2b shows a TLM composed of purely resistive z_1 and z_3 , and purely capacitive z_2 . In this model, the upper and lower rails consist of a single resistance element (r_i and r_e), which accounts for the resistance that a Li^+ experiences across solid electrolytes and an electron experiences across active materials and conductive additives, respectively. The CPE element at the crosswise rail (cpe_{int}) represents the double-layer capacitance at the electrolyte/active material interface. The resulting impedance plot consists of a straight line inclined at 45° with respect to the real axis in the high-frequency region, followed by a semicircle in the low-frequency region (referred to as the Gerischer behavior) [33]. The magnitude of r_i affects the overall sizes of the Gerischer-shaped impedance spectra, whereas the variation in r_e shifts the location of the low-frequency limits (Fig. 2c).

In addition to pure resistance, other factors can affect transport processes in composite electrodes, such as grain boundaries within solid electrolytes and contacts between electronically conductive particles. To be specific, the “grain-boundary” contribution refers to the impedance across inter-grain regions, whereas the “bulk” contribution denotes the intra-grain transport. To examine the impact of these factors, TLMs were constructed with each rail consisting of a series connection of a resistance element and a parallel $r_{gb} || cpe_{gb}$ element, representing the bulk resistance and grain-boundary impedance, respectively (Fig. 2d–g). Parallel $r_{gb} || cpe_{gb}$ elements were added to z_1 and z_3 . For both cases,

additional semicircles appear in the high-frequency region (Fig. 2d and f). Furthermore, the ratio of the bulk ionic resistance $r_{i,bulk}$ to the ionic grain-boundary resistance $r_{i,gb}$ was varied with a fixed total ionic resistance. The size of the semicircle at the high-frequency region increases as $r_{i,gb}$ increases (Fig. 2e). The local minimum between the two semicircles shifts to a lower frequency as the ionic grain-boundary capacitance $c_{i,gb}$ increases. Similar behavior is observed with varying electronic grain-boundary resistance $r_{e,gb}$ and capacitance $c_{e,gb}$ (Fig. 2g). The detailed parameters used in simulations are provided in Table S1.

Using the same modeling approach, equivalent circuit modeling was performed for TLMs with the electron blocking–ion blocking condition, which represent a composite electrode in a full cell where charge transfer occurs (Fig. 3a). Fig. 3b depicts the TLM with purely resistive z_1 and purely capacitive z_2 . The resulting impedance spectra consist of a straight line inclined at 45° with respect to the real axis in the high-frequency region and a capacitive line in the low-frequency region. The charge transfer resistance r_{int} is incorporated into the interfacial impedance z_2 , forming a parallel connection with cpe_{int} (Fig. 3c). In the resulting Nyquist plot, the high-frequency line with a 45° phase angle is ascribed to the ionic transport impedance, whereas the low-frequency semicircle is assigned to the interfacial impedance.

To incorporate solid-state diffusion of Li into the TLM, the generalized finite-length Warburg element z_w was introduced in the crosswise rail (Fig. 3d) [34]. The Warburg exponent, α , which corresponds to the degree of frequency dispersion for diffusion, was set to 0.5. In the low-frequency region, the Nyquist plot exhibits a straight line inclined at a 45° phase angle with respect to the real axis, followed by a perpendicular line as the frequency decreases, reflecting the Warburg-to-capacitive transition [34,35]. These TLMs without z_3 elements, where the lower rail acts as a non-blocking wire, represent composite electrodes with high electronic conductivity (Fig. 3b–d). In contrast, Fig. 3e shows the TLM with electronic resistance, which results in an overall impedance shift toward higher values along the real axis. The influence of the resistive elements was further investigated by varying the magnitude of r_i and r_{int} (Fig. 3f and g). The resultant Nyquist plots exhibit an elongated diagonal region with increased r_i and an enlarged semicircle with increased r_{int} . These parametric simulations

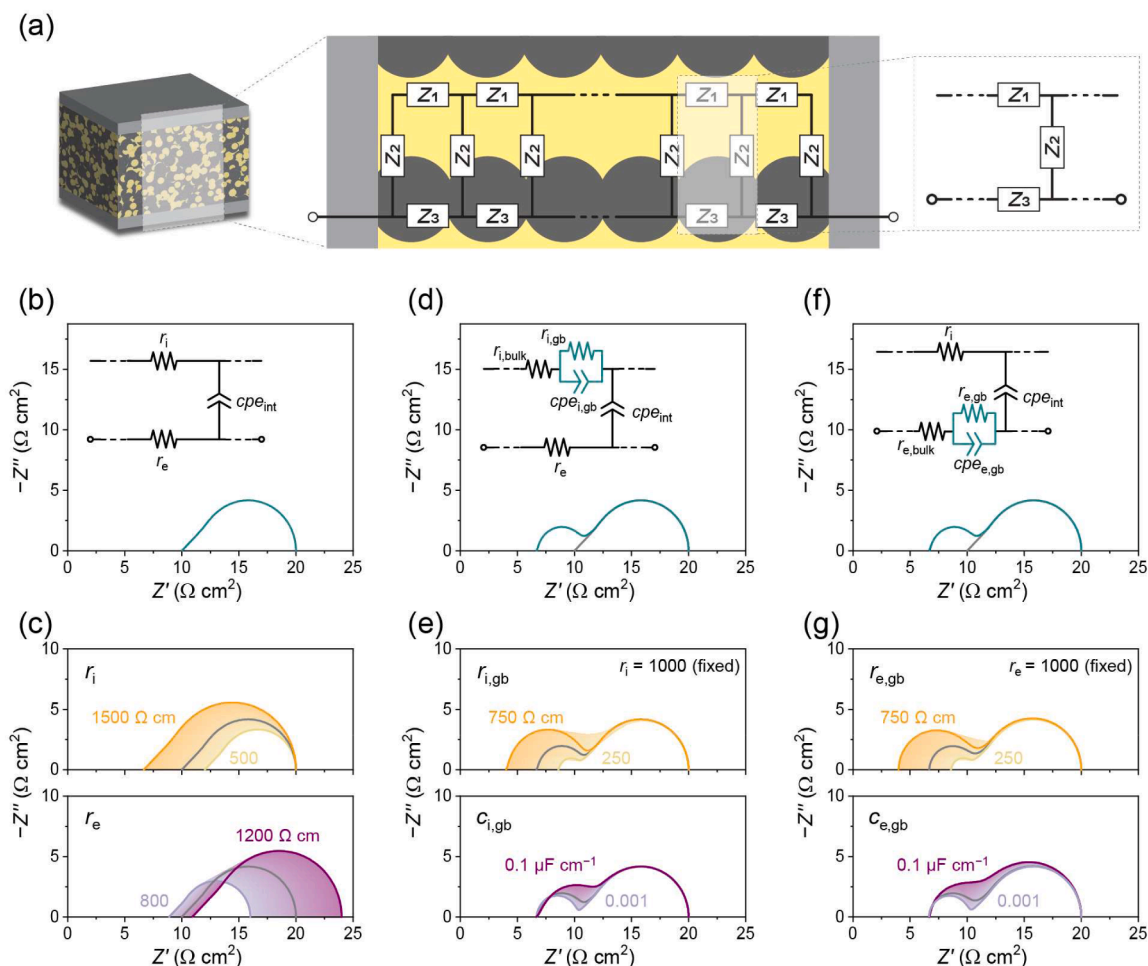


Fig. 2. (a) TLM of the ion blocking–ion blocking condition and its circuit elements. (b) TLM configuration with purely resistive components and resulting impedance spectrum. (c) Parametric study with variations in r_i and r_e . (d) TLM configuration with ionic grain-boundary components and resulting impedance spectrum. (e) Parametric study with variations in $r_{i,gb}$ and $c_{i,gb}$. (f) TLM configuration with electronic grain-boundary components and resulting impedance spectrum. (g) Parametric study with variations in $r_{e,gb}$ and $c_{e,gb}$. Detailed parameters for simulations can be found in Table S1.

demonstrated the effect of each element on the specific regions of impedance spectra. Table S2 provides the detailed parameters employed in the simulation.

3.2. Model cell experiments and EIS analyses

Two types of model cells, i.e., composite electrode symmetric cells and three-electrode full cells, were fabricated with various electrode formulations to demonstrate quantitative analyses of composite electrodes based on the established TLMs. Their impedance responses were measured at varying temperatures. LPSCl, NCM811, and Super P were employed as the solid electrolyte, active material, and conductive additive, respectively. SUS foils were employed as current collectors to minimize variations in contact resistance, owing to their wide electrochemical stability [36]. The composition of the active material was set to 62, 72, and 82 wt %, while the conductive additive content was fixed at 1 wt %. Here, it should be noted that the NCM811 active material was coated with $LiNbO_3$ (LNO) to suppress interfacial chemical degradation reactions [37,38]. After impedance measurements, each symmetric cell was disassembled, and the composite electrode thickness (L) was measured under inert conditions. The L values of the full cells were calculated from those of the symmetric cells, considering the mass ratio of the composite cathode mixtures. Distribution of relaxation time (DRT) analysis was also conducted to deconvolute time constants and determine the appropriate equivalent circuit model [39].

The impedance spectra of the symmetric cells, consisting of a composite electrode sandwiched between two ion-blocking SUS foils, are shown as Nyquist plots and DRT spectra in Fig. 4a–d. Similar to the simulation results presented in Fig. 2d and f, the Nyquist plots exhibited two depressed semicircles (Fig. 4a). With increasing NCM811 content, the size of the spectrum decreased, and the real axis intercept at the low-frequency limit shifted toward a lower value. The DRT spectra of the symmetric cells displayed five polarization peaks, and their overall peak intensities drastically decreased as the NCM811 content increased (Fig. 4b). Both TLMs with ionic and electronic $r_{gb} \parallel cpe_{gb}$ elements (Fig. 2d and f) can provide a good interpretation of these experimental results.

To clarify the origins of the impedance in the high-frequency region, the cell systems were investigated at varying temperatures (Fig. 4c and d). The cathode composition was fixed at the NCM811:LPSCl:Super P ratio of 62:27:1, and the temperature increased with a step size of 15 °C. As the temperature rose from 30 to 60 °C, the radius of the high-frequency semicircle decreased (Fig. 4c), accompanied by a reduction in the intensity of the DRT peaks in the high frequency region (Fig. 4d). These temperature-dependent shifts indicate that the two DRT peaks in the high-frequency region are related to ion transport within grain crystallites and across grain boundaries. Therefore, the high-frequency semicircle can be attributed to the ionic grain-boundary impedance.

The measured impedance spectra of the symmetric cells were quantitatively analyzed using the TLM incorporating ionic grain-

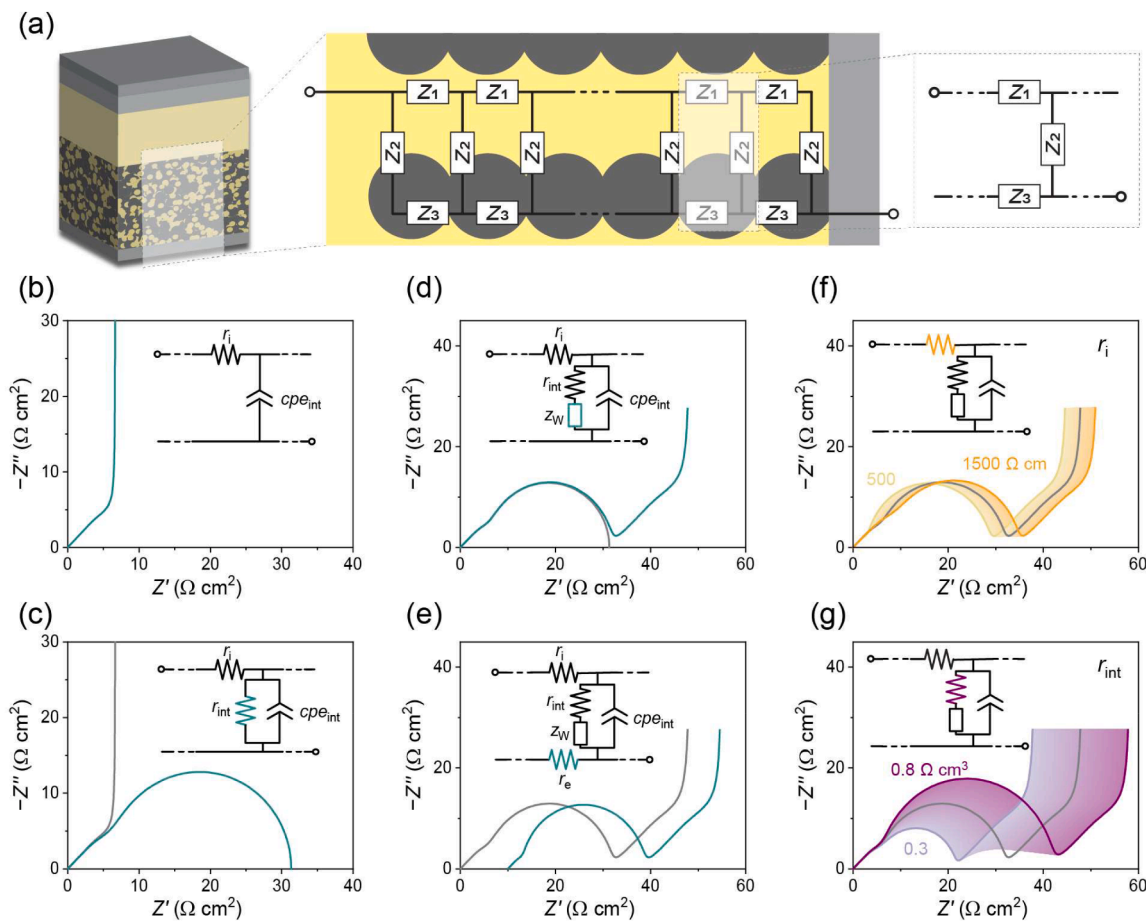


Fig. 3. (a) TLM for the electron blocking-ion blocking condition and its circuit elements. TLM configurations for the simulation and impedance spectra with the components of (b) cpe_{int} , (c) r_{int} , (d) z_w , and (e) r_e . Parametric study with variations in (f) r_i and (g) r_{int} . The corresponding TLMs are depicted in the inset. Detailed parameters for simulations can be found in Table S2.

boundary impedance (Fig. 4e and f, and Table S3). Except for the cell with 62 wt % NCM811, electronic resistances were negligible compared with the ionic resistances, and the ionic resistance decreased with increasing temperature. Although higher SE content is generally expected to reduce ionic resistance, the composite electrode with 82 wt % NCM811 exhibited the lowest ionic resistance, followed by those with 62 and 72 wt %. This trend can be attributed to differences in densification processes. Since the same powder mass was used for all compositions, a higher NCM811 fraction led to a smaller total volume, resulting in thinner and more compact electrodes under the same pressure, which likely enhanced the percolation of ionic pathways [40,41]. Notably, the ionic resistance at the grain boundaries contributed significantly, with a magnitude comparable to that of the bulk resistance. Such a high grain-boundary contribution is consistent with previously reported kinetic characteristics of sulfide electrolytes [42,43].

For the analysis of the interfacial charge transfer, cells with the electron blocking-ion blocking boundary condition were fabricated. To isolate the signal originating solely from the composite electrode, EIS measurements were performed using three-electrode full cells. The three-electrode cells consisted of a Li-In anode (counter electrode), LPSCl solid electrolyte, composite cathode mixture (working electrode), Li-Ag alloy wire (reference electrode), and SUS current collectors [44]. The impedance measurements were conducted under the same temperature and composition conditions as in the symmetric cell experiments. The impedance profiles and corresponding DRT spectra of three-electrode full cells are shown in Fig. 5a–d. The Nyquist plots exhibited a straight line inclined at 45° with respect to the real axis in the high-frequency region, a single depressed semicircle in the

mid-frequency range, and another straight line in the low-frequency region (Fig. 5a and c). Fig. 5b and d show the corresponding DRT spectra. The intensity of the second-largest DRT peak (0.1–1 kHz) decreased with increasing temperature and active material content. Thus, the semicircle observed in this mid-frequency range was attributed to the interfacial charge transfer impedance, whereas the inclined line in the high-frequency region was assigned to the ionic transport contribution. The straight line in the low-frequency region was ascribed to the solid-state diffusion process.

Fig. 5e depicts the TLM corresponding to the composite electrodes in three-electrode full cells. For impedance fitting, an equivalent circuit consisting of a series connection of the TLM and a parallel $R \parallel CPE$ (contact) element was employed. Here, the $R \parallel CPE$ (contact) element represented the contact impedance of the cell. The responses in the high-frequency region were attributed to both the parallel $R \parallel CPE$ elements for contact and the ionic pathway. The charge transport characteristics obtained from the symmetric cell analysis were used for the fitting of the full cell. In particular, the ionic transport and electronic transport resistances were assumed to be identical for both the symmetric and full cells. The ionic transport resistance of the symmetric cell was taken as the sum of $r_{i,bulk}$ and $r_{i,gb}$. The fitting results of three-electrode full cells are summarized in Fig. 5f and Table S4. The charge transfer resistance decreased with increasing temperature due to enhanced kinetics. Moreover, the cell with 82 wt % NCM811 exhibited the lowest charge transfer resistance among the compositions. This result is attributed to an enlarged electroactive surface area between the active material and the solid electrolyte, resulting from the higher NCM811 content.

Among multiple composite cathode components, carbon black

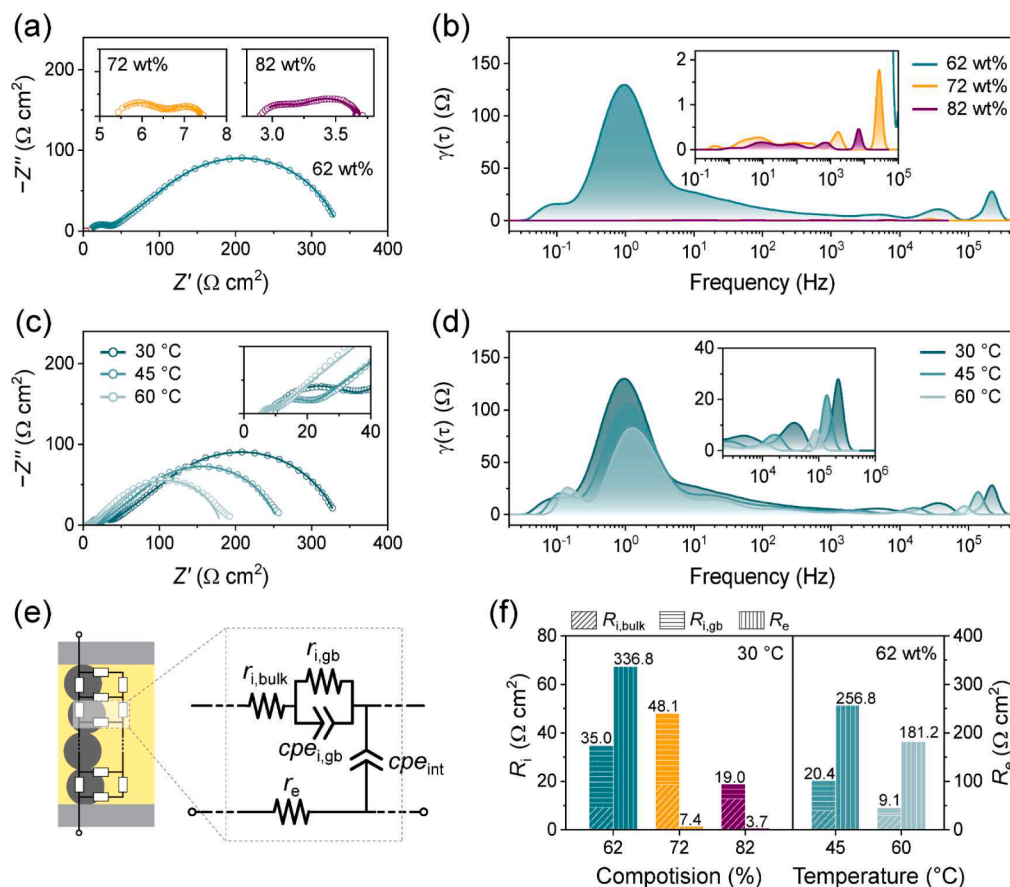


Fig. 4. Impedance spectra of the symmetric cells measured with the NCM composition of 62, 72, and 82 wt %, presented as (a) Nyquist plots and (b) DRT spectra. Impedance spectra of the symmetric cells measured at temperatures of 30, 45, and 60 °C, presented as (c) Nyquist plots and (d) DRT spectra. (e) TLM used for the fitting and (f) corresponding quantification results.

(Super P) was employed as conductive additives, as it is commonly used to provide electronic percolation in conventional ASSBs. However, trace metallic impurities are introduced during manufacturing of carbon black, which can hinder the stable cycling of ASSBs [45]. In addition, minimizing the interfacial contact between the sulfide electrolyte and conductive additive is critical to suppress the electrolyte decomposition reaction. To address these issues, one-dimensional (1D) conductive carbon materials have been investigated as alternatives [46]. In this respect, we performed additional experiments using vapor-grown carbon fiber (VGCF) as the conductive additives (Fig. S1). Symmetric and full cells were fabricated with the cathode composition was fixed at the NCM811:LPSCI:VGCF ratio of 62:27:1. The symmetric cell employing VGCF exhibited considerably lower electronic resistance than its Super P counterparts with identical composition, indicating that the 1D carbon framework provides more efficient electronic percolation than carbon black (Fig. S1a,b). Moreover, the impedance responses of the full cell employing VGCF showed decreased interfacial resistance (Fig. S1c,d). These results suggest that the 1D VGCF not only enhanced electronic percolation but also reduced the interfacial area between conductive additives and sulfide electrolytes, thereby mitigating oxidative decomposition of electrolytes.

3.3. Interplay between interfacial and ionic transport kinetics

To investigate the interplay between interfacial and ionic transport kinetics, we evaluated the impedance spectra of model cells employing uncoated NCM811 as the active material, instead of LNO-coated NCM811. The Nyquist plots of symmetric cells exhibited two clearly separated semicircles, reflecting a significant grain-boundary

contribution to ionic transport (Fig. 6a). This grain-boundary contribution may result from sluggish ionic transport at the surface of the uncoated active material. As the active material content increased, the electronic resistance decreased while the ionic resistance increased. Temperature-variation experiments were also conducted at a fixed NCM811 content of 72 wt %, in which the electronic and ionic resistances followed trends similar to those observed in cells using LNO-coated active materials (Fig. 6b and Table S5).

More importantly, the interfacial charge transfer kinetics of the uncoated active materials were evaluated using ion blocking–electron blocking model cells (Fig. 6c and d). For the uncoated active material, the semicircle in the mid-frequency region attributed to interfacial charge transfer was significantly larger than that of coated NCM811 (Fig. 6c). The oxidative degradation of uncoated NCM811 with LPSCI resulted in large interfacial charge transfer resistance (i.e., sluggish kinetics) (Fig. 6d and Table S6), whereas the LNO protective layer suppressed side reactions at the interfaces between the solid electrolytes and active materials (Fig. 5a and c) [47]. For the coated active materials, the ionic transport region (high-frequency line with a 45° phase angle) was nearly indistinguishable from the charge transfer region (mid-frequency semicircle) in the impedance spectra. In contrast, for the uncoated active materials, the substantial interfacial charge transfer region could be clearly distinguished from the ionic transport region. This distinction enabled the impedance to be readily decoupled into ionic transport and charge transfer contributions, thereby allowing direct evaluation of charge transfer properties without additional analysis of charge carrier transport.

To further elucidate the influence of the relative magnitudes of ionic transport and charge transfer kinetics on the impedance spectra,

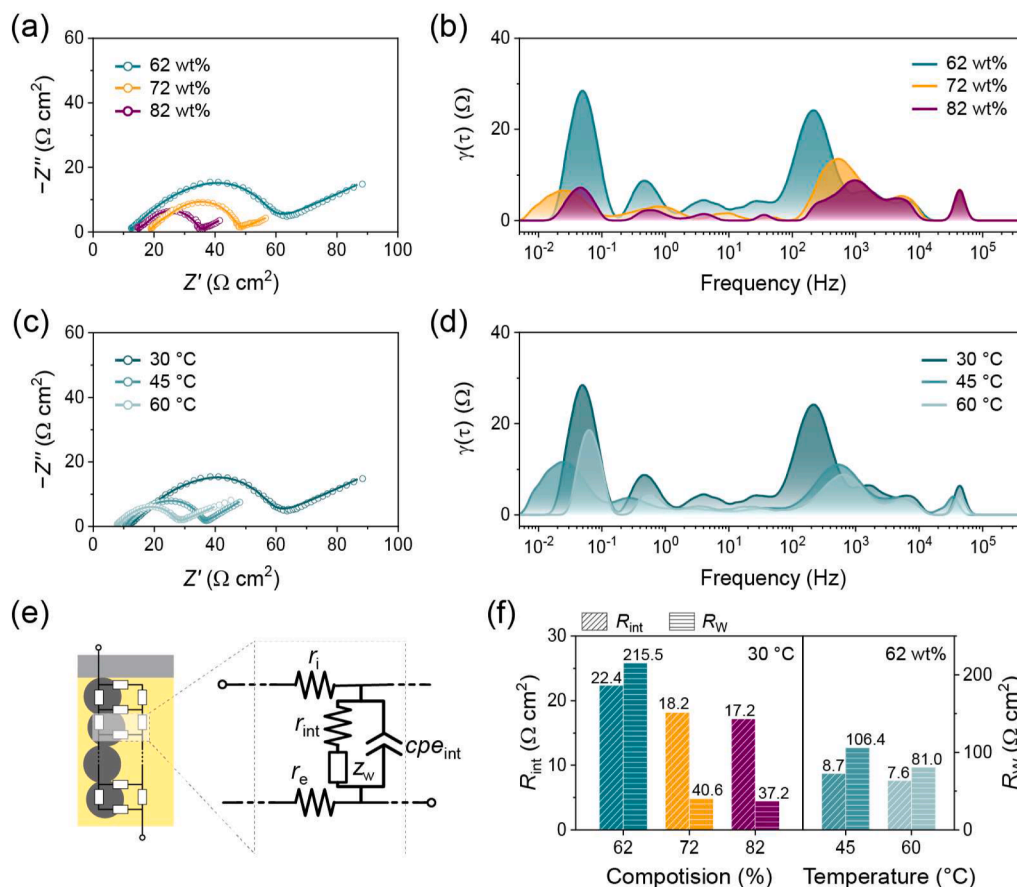


Fig. 5. Impedance spectra of full cells measured with the NCM composition of 62, 72, and 82 wt %, presented as (a) Nyquist plots and (b) DRT spectra. Impedance spectra of full cells measured at temperatures of 30, 45, and 60 °C, presented as (c) Nyquist plots and (d) DRT spectra. (e) TLM used for the fitting and (f) corresponding quantification results.

additional equivalent circuit modeling was conducted. Morasch et al. previously investigated how the ratio of charge transfer resistance to pore ionic resistance, which depends on the areal capacity, affects impedance responses of porous electrodes in Li-ion battery systems [48]. Using a similar approach, we performed a parametric study with the TLM by varying the ratio R_{int}/R_i (Fig. 6e). Detailed model parameters are provided in Table S7. When R_{int}/R_i ratio was high (3 and 1), the ionic transport and charge transfer regions of the impedance spectra were clearly distinguishable, resembling those observed for the uncoated active materials. In contrast, when R_{int}/R_i was low (1/4 and 1/16), the two regions overlapped and became indistinguishable. Furthermore, the resulting impedance responses were nearly identical for $R_{int}/R_i = 1/4$ and 1/16. Noticeable differences appeared only in the low-frequency diffusion region, arising from the distinct values of r_{int} . These simulation results suggest that unless charge transfer kinetics are appreciably slower than ionic transport, impedance analysis in full cell configurations can lead to misleading interpretations. For such full cell impedance responses, independent quantification of the charge transport resistance is required for the accurate evaluation of kinetic processes.

Finally, based on the TLM-based impedance decoupling framework, we extended our analysis to a halide electrolyte system. Symmetric and full cells were fabricated using the uncoated NCM811 as the active material and LZC as the solid electrolyte, which has lower ionic conductivity (0.51 mS cm^{-1}) than that of LPSCI (1.6 mS cm^{-1}) [49,50]. The electrode composition was fixed at 72:27:1 (NCM811:LZC:Super P), and the measured thickness L of the symmetric cell was $170 \mu\text{m}$. The impedance spectrum of the symmetric cell using the LZC electrolyte exhibited significant grain-boundary contributions (Fig. 6f). As expected, ionic transport resistance was higher than that of the

LPSCI-based counterpart with the same composition. Therefore, ionic transport and charge transfer regions were indistinguishable in the full cell measurements (Fig. 6g). This is likely due to the combination of low charge transfer resistance and high ionic transport resistance, as predicted by the simulations shown in Fig. 6e. To accurately deconvolute the kinetic processes in the full cell, the ionic and electronic transport resistances quantified from the symmetric cell were used to fit the impedance response. It turned out that the resulting R_{int} value was significantly lower than that of the LPSCI-based system, indicating reduced interfacial oxidative degradation (Fig. 6d). These results are consistent with the reported electrochemical properties of halide electrolytes, which offer higher oxidative stability but lower ionic conductivity than sulfides [12,50]. Thus, the developed TLM framework enabled reliable impedance decoupling in the halide-based system, particularly through complementary analysis of charge transport characteristics using symmetric cells.

4. Conclusion

In this study, we quantitatively analyzed electrochemical processes in composite electrodes for ASSBs using modified TLMs composed of distributed impedance elements. Model simulations of various TLM configurations, combined with impedance spectra of symmetric and full cells, enabled systematic decoupling of electrochemical processes. Charge carrier transport kinetics were evaluated with the ion blocking-ion blocking TLM. While equivalent circuit modeling suggested that a high-frequency semicircle could originate from either electronic contact or ionic grain-boundary resistances, temperature-dependent experiments indicated that it was mainly caused by ionic grain-boundary

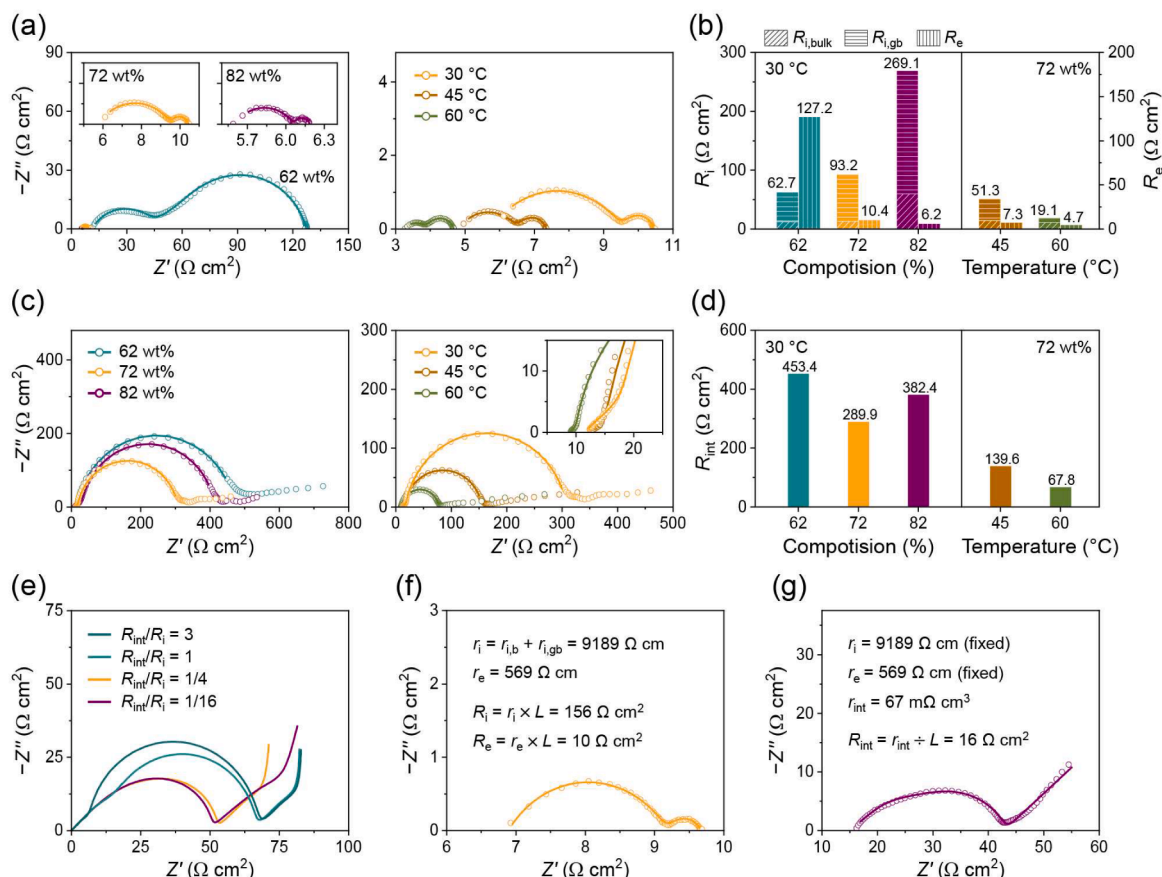


Fig. 6. (a) Impedance spectra of symmetric cells employing uncoated NCM811 and (b) corresponding quantification results. (c) Impedance spectra of full cells employing uncoated NCM811 and (d) corresponding quantification results. (e) Simulated impedance spectra with varied R_{int}/R_i values. Impedance spectra of the composite electrode with halide electrolyte measured obtained from (f) symmetric and (g) full cell.

impedances between solid electrolyte particles. Intensified pellet densification improved ionic percolation pathways, while higher active material content rendered electronic transport resistance negligible. Interfacial charge transfer kinetics were analyzed using the electron blocking-ion blocking TLM in addition to solid-state diffusion within the active materials. Electrode composition dictated the electrochemically active surface area and controlled the charge transfer resistance. The interplay between charge carrier transport and interfacial charge transfer kinetics was further investigated using uncoated NCM811 and halide electrolytes. At low ionic conductivity and/or high charge transfer rate, the ionic transport and interfacial charge transfer regions in the impedance spectra became indistinguishable. Consequently, independent decoupling based solely on full cell measurements can be misleading, and complementary analysis with symmetric cells is required for accurate evaluation. This study establishes a robust strategy for the quantitative analysis of multiple reaction kinetics in ASSBs.

CRediT authorship contribution statement

Siwon Kim: Writing – original draft, Investigation, Formal analysis.
Junhee Kang: Investigation. **Jae Hyun Park:** Investigation. **Jong-Won Lee:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.electacta.2025.147413](https://doi.org/10.1016/j.electacta.2025.147413).

Data availability

Data will be made available on request.

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